Remote Development Project Report

Student Name: 徐兆冉
Student ID: ZY2557210

System Configuration

Item	Information
CPU Model	AMD Ryzen 7 H 255 w/ Radeon 780M Graphics, 16 cores, x86_64
Memory Size	15 Gi total, 620 Mi used, 14 Gi free
Operating System Version	Linux localhost 6.6.87.2-microsoft-standard-WSL2
Compiler Version	gcc (Ubuntu 13.3.0-6ubuntu2~24.04.1) 13.3.0
Python Version	Python 3.13.12

Implementation Details

Python Language Implementation

Source Code

def matrix_multiply(A, B): # 维度检查 if len(A[0]) != len(B): raise ValueError("矩阵维度不匹配") # 初始化结果矩阵 result = [[0.0] * len(B[0]) for _ in range(len(A))] # 三重循环 for i in range(len(A)): for j in range(len(B[0])): for k in range(len(B)): result[i][j] += A[i][k] * B[k][j] return result

if **name** == "**main**": # 示例矩阵 A (2x3) A = [[1, 2], [3, 4]] # 示例矩阵 B (3x2) B = [[5, 6], [7, 8]] C = matrix_multiply(A, B) print("矩阵 A:") for row in A: print(' '.join(map(str, row))) print("\n矩阵 B:") for row in B: print(' '.join(map(str, row))) print("\n矩阵 A * B:") for row in C: print(' '.join(map(str, row)))

Execution Command

```
python assignment.py
## Algorithm Verification

### Correctness

To verify the correctness of the matrix multiplication algorithm, a small 2×2 test case was used.

The input matrices were:

\[
A =
\begin{bmatrix}
1 & 2 \\
```

```
3 & 4
\end{bmatrix}
\]

\[
B =
\begin{bmatrix}
5 & 6 \\
7 & 8
\end{bmatrix}
\]

The expected result by manual calculation is:

\[
A \times B =
\begin{bmatrix}
19 & 22 \\
43 & 50
\end{bmatrix}
\]

## Conclusion

In this project, I learned how to use Unix/Linux command line tools to collect
system information and manage files. I also practiced writing technical
documentation in Markdown format and implemented matrix multiplication in Python.
By testing the program with a small example and comparing the output with manual
calculation, I verified the correctness of the algorithm. This project helped me
better understand remote development workflows and the differences between
documentation and implementation tasks.
The output of the Python program matched the manually calculated result exactly.
Therefore, the algorithm was verified to be correct for this test case.

## References

1. Python Official Documentation
2. GCC Official Documentation
3. Linux manual pages
4. Markdown Guide
```